

When considering the categories associated with the normative score ranges, the professional should keep in mind that the focus of interpretation is the respondent's performance; that is, it is the respondent's performance on the RCFT, rather than the respondent per se, that may be deficient or impaired. This distinction is especially important when evaluating older adults because of the greater variability in RCFT performance found among neurologically intact subjects in this age range.

INTERPRETATION OF RCFT PERFORMANCE

Performance on Single RCFT Measures

Results of validity analyses indicate that the Copy score measures visuospatial constructional skills. Lower Copy scores indicate reduced visual-perceptual and visuomotor integration skills. In the factor analyses with both normal subjects and brain-damaged patients, Time to Copy and Copy loaded on different factors, suggesting that these two variables measure different constructs. In the absence of perceptual or motor difficulties, higher Time to Copy scores indicate reduced speed of cognitive processing. Reduced Immediate and Delayed Recall scores suggest reduced visuospatial recall ability, while Recognition Total Correct measures the ability to retrieve visuospatial material when given retrieval cues. Meyers and Lange (1994) found that the Recognition trial measures a unique dimension of visuospatial memory functioning that was not tapped by the Immediate and Delayed Recall trials.

Memory Profile Patterns

One interpretive approach to RCFT scores is to evaluate the profile or pattern of scores on Immediate Recall, Delayed Recall, and Recognition Total Correct. Chapter 6 discusses the results of research conducted with brain-damaged patients on the diagnostic utility of these memory profile patterns. This interpretive approach is similar to the codetype approach used to interpret multiscale personality inventories (Greene, 1991) and is based on the theoretical discussion of memory performance by Sohlberg and Mateer (1989). Prior to profiling, RCFT

raw scores should be converted to age-corrected *T* scores using the normative tables in Appendix C.

Profiles are judged to reflect a **normal** pattern of performance if scores on Immediate Recall and Delayed Recall are above a *T* score of 40 and exhibit little or no slope. In normal memory patterns, the Delayed Recall score may be slightly higher than the Immediate Recall. A profile is judged to reflect disrupted attention if all three *T* scores are below 25 and exhibit little or no slope. The **attention** profile represents very impaired performance on all three RCFT memory measures. An **encoding** pattern is inferred if the Immediate and Delayed Recall *T* scores are below a *T* score of 25 for respondents 75 years and older or a *T* score of 23 for respondents younger than 75 years and are generally equivalent (i.e., within 3 *T*-score points). Recognition Total Correct should be no more than 10 *T*-score points higher than the recall scores. The encoding pattern reflects negligible improvement in memory when given retrieval cues.

A profile is judged to reflect disrupted **storage** processes when the profile slopes downward and to the right, with Immediate Recall higher than Delayed Recall and Delayed Recall higher than Recognition Total Correct. A **retrieval** pattern is inferred if the Immediate Recall and Recognition Total Correct scores are generally equivalent, often in the normal range, and the Delayed Recall score is less than either the Immediate Recall or the Recognition Total Correct score, forming a "V" shape. A retrieval pattern is also inferred when the Immediate Recall and Delayed Recall *T* scores are generally equivalent, the Recognition Total Correct *T* score is higher than both the Immediate Recall and Delayed Recall *T* scores, and the pattern does not qualify as an encoding profile. The memory profile patterns for a sample of 100 brain-damaged patients were rated by the senior author. Figure 2 illustrates the mean RCFT score for each profile pattern in this sample. Refer to chapter 6 for additional information about the derivation and empirical validation of the memory profile patterns.

Qualitative Aspects of Performance

In both normal and brain-damaged samples, several items from the Recognition trial (Items 1, 4, 6, 10, 11, 18, and 21) are rarely answered incorrectly. An incorrect answer on these items is referred to as an

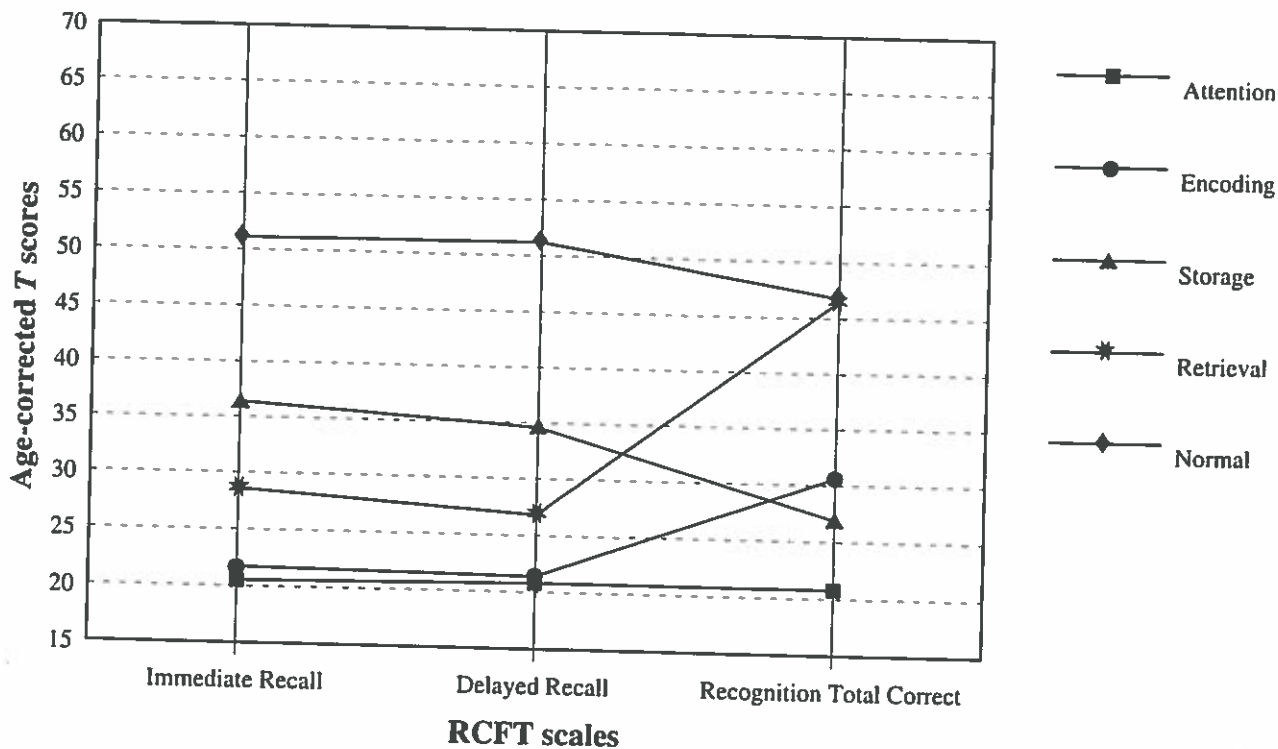


Figure 2. Typical RCFT memory profile patterns.

Atypical Recognition Error (Knight & Meyers, 1995). In both normal and brain-damaged samples, the accurate drawing of an individual scoring unit (although not necessarily correctly placed) at the Delayed Recall trial is highly predictive of a correct answer to the corresponding Recognition trial item. Stated differently, a respondent is highly likely to recognize a scoring unit if he or she correctly recalled it at the Delayed Recall trial. A Recognition Failure Error refers to the situation where a respondent correctly recalls (i.e., draws) a scoring unit at the Delayed Recall trial but subsequently fails to recognize the unit during the Recognition trial.

Normal subjects rarely produce even one Atypical Recognition Error. However, both brain-damaged patients functioning below Rancho Los Amigos Level VIII (Hagan, Malkmus, Durham, & Bowman, 1979) and experimental dissimulators (normal subjects instructed to malingering) produce a similar number of Atypical Recognition Errors, ranging from zero to five or more. Patients with mild head injury

do not produce Atypical Recognition Errors or Recognition Failure Errors (Knight & Meyers, 1995). For respondents at a Rancho Los Amigos Level of VIII, a cutoff score of 2 for Atypical Recognition Errors and Recognition Failure Errors is cautiously suggested to identify cases of questionable motivation needing further investigation for possible response styles. As these cutoff scores have not been adequately cross validated with independent samples of patients with confirmed neurobehavioral and cognitive impairment, appropriate caution should be exercised in interpreting elevations on these scores. For example, an individual who is functioning below an independent level of daily living skills may achieve a score that suggests questionable motivation or dissimulation, which may, in fact, be the result of confusion and disorientation. However, for an individual who is functioning independently, the presence of two or more Atypical Recognition Errors or two or more Recognition Failure Errors would raise a question regarding motivation (Knight & Meyers, 1995).